Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/07/2008
PI Name:	Doerr, Harold K. M.D.		
Project Title:	Medical Operations Support Team		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hdoerr@bcm.edu	**Fax:**	FY 713-798-7345
PI Organization Type:	UNIVERSITY	**Phone:**	713-798-7356
Organization Name:	Baylor College of Medicine		
PI Address 1:	1709 Dryden Rd		
PI Address 2:	Suite 1700		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/01/2007	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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NASA has understood the educational benefits of using simulation during its history as demonstrated by its integration of simulation into multiple aviation disciplines to prepare its astronauts for space flight. The NASA Space Life Sciences Directorate (SLSD) at the Johnson Space Center (JSC) has observed these benefits along with recognizing the recent success of medical patient simulation in teaching terrestrial medicine. As a result, the SLSD has developed an infrastructure to use high-fidelity medical patient simulation as a teaching tool for astronaut crew medical officers (CMO) and flight surgeons (FS). In order to integrate medical patient simulation into medical training programs, the SLSD tasked the NSBRI with constructing a team that has the expertise and experience to develop medical patient simulation-based curricula.

The NSBRI assembled diverse clinical and scientific personnel from both inside and outside the space medicine community to create the Medical Operation Support Team (MOST). Since its inception, the team has used its unique skill sets to develop eight (8) medical patient simulation-based products for these cohorts per the direction of NASA

Task Description:	Medical Operations Branch (MOB), the Astronaut Office, and the NASA Human Research Program (HRP). Overall, the MOST has used its expertise with their strong space medicine collaborations and NASA-JSC's medical patient simulator infrastructure to develop robust space medicine training products. Goals for '2007-'08: Goal 1: Adapt Human Patient Simulator (HPS) physiologic scripting and modeling to space flight training. Goal 2: Develop an integrated, multimedia medical curriculum and web development to integrate with the HPS training sessions. Goal 3: Develop, evaluate, and validate space medical procedures. Goal 4: Maintain and expand the multidisciplinary team in conjunction with NASA-JSC's needs and requests. The key findings of these goals and their impact are listed below. Please note that the following tasks were accomplished on a (mostly) volunteer/non-funded basis due to the decrease in project funding: Goal 1 -The MOST developed an experimental integrated medical simulation program for the flight surgeons and astronaut crew medical officers (CMO) of the STS-118 mission. This came at the direct request of physician-astronaut, Dr. Dave Williams and the flight surgeons for STS-118. The session was the first simulation exercise of its kind and is being reviewed by the MOB for possible integration into the Space Medicine Training flow. IMPACT: The impact of this training has been positive as demonstrated by the CMOs and flight surgeons from STS-125 requesting similar training. - The MOST developed an emergency medicine scenario for an 'Emergency Medical Drills' session that was conducted during the 12th and 13th expeditions of the NASA Extreme Mission Operation (NEEMO) project. This work was tasked by the NASA-JSC Medical Operations Manager and was reviewed/approved by the NASA-JSC Human Research Program (HRP). IMPACT: The impact of this training has been positive as demonstrated by 1) the positive feedback from NEEMO crew members and 2) the NEEMO coordination team requesting to conduct similar training for future NEEMO missions, in particular, NEEMO 14. -The team expanded the "Clinical Core Competency Training" for the NASA-JSC Flight Surgeons by adding and teaching an Airway Management lesson (Airway Management #1). In addition, the team drafted a follow-up Airway Management lesson (Airway Management #2) with the objectives of using simulation-based instruction to teach the FS cohort how to use the airway management techniques from the previous lesson in real-time clinical scenarios relating to both terrestrial and space flight environments. These 2 lessons result in a total of 3 Core Competency Training Sessions developed by the MOST for the NASA-JSC Flight Surgeons. IMPACT: The impact of these sessions has been positive as the NASA-JSC Medical Operations Manager and the NASA-JSC Space Medicine Training Lead have indicated that they would like the MOST to continue developing and teaching more sessions. -The MOST continued to provide "Post-Soyuz Landing Clinical Training" (Medical simulation training specific for post-International Space Station (ISS) Soyuz landings in Kazakhstan was given to two teams prior to landings) and "Individual Clinical Refresher Training" (Individual clinical training using medical patient simulation that is designed to strengthen a specific clinical skill set) on a per-request basis from the MOB's NASA-JSC Flight Surgeons. IMPACT: The impact of these training sessions has been positive as demonstrated by the positive feedback from MOB participants. Goal 2 The frame work of the MOST website prepares the student for their session by having the student take a brief pre-session test. Doing so identifies the student's weaknesses, thus allowing the session to be oriented to help strengthen the student's weaknesses. The data collected from pre- and post-session tests allows the MOST to measure the effectiveness of their instruction. The website infrastructure was finalized and was ready to be populated with learning modules; however, the decrease in funding halted next steps. IMPACT: minimal Goal 3 Although the MOST has received positive feedback from the NASA-JSC MOB regarding the team's evaluations of procedures and equipment throughout the project, the decrease in project funding minimized the time availability to this proposed goal, thus preventing the team from conducting any formal evaluations of procedures and equipment. IMPACT: minimal Goal 4 The size of the MOST was maintained for this research year; however, decrease in project funding prevented the team from expanding to fully meet the needs requested from NASA-JSC Space Medicine. IMPACT: minimal
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	PROJECT IMPACT: The operational and environmental constraints imposed by space flight demand novel approaches to mitigating medical issues. The MOST's continued use of medical patient simulation to develop space-relevant clinical training products for the NASA Flight Surgeon cohort would provide a significant advance in telemedicine capability within NASA Mission Control for both LEO and lunar class missions. Terrestrial benefits from this project are also anticipated including the adaptation of the training for providing medical care in remote environments with limited medical resources such as the wilderness, underwater habitats, high-altitude base-camps, and polar expedition locales. FS Impact: CME credits

	The Medical Operation Support Team (MOST) developed an experimental integrated medical simulation program to instruct astronauts and their ground-based flight surgeons how to mitigate medical issues related to space flight. The team conducted this experimental training for two cohorts: Training was developed for the flight surgeons and astronaut crew medical officers (CMO) of the STS-118 Space Shuttle mission, per the request of the Lead CMO. The team developed curricula using medical patient simulators to train the CMOs how to mitigate space flight-related medical events while being remotely guided by their ground-based flight surgeons in Mission Control. The session was the first simulation exercise of its kind and is being reviewed by the Medical Operations Branch (MOB) at NASA-Johnson Space Center for possible integration into the Space Medicine Training flow.
Task Progress:	The MOST developed an emergency medicine scenario for an 'Emergency Medical Drills' session that was conducted during the 12th and 13th expeditions of the NASA Extreme Mission Operation (NEEMO) project. NEEMO consists of an underwater habitat that is supposed to mimic the isolation and constraints experience during space flight and, thus, this project acts as a good test-bed for researchers to evaluate procedures and equipment for possible use during space missions. The data collected by the MOST from these 'Drills' demonstrated that that having a physician as a CMO led to an expeditious identification and treatment of an unknown medical condition. As a result of this finding, the Medical Operations Branch (MOB) at NASA-Johnson Space Center is reviewing whether implementation of physician-astronauts as Lead CMO would be beneficial for long duration space missions (e.g. International Space Station, Moon, and Mars).
Bibliography Type:	Description: (Last Updated: 10/22/2008)
Abstracts for Journals and Proceedings	Doerr HK, Hurst V, Williams D, Morgan BR,Tadeo T, Powers W, Hughlett J, Marshburn T, Polk J, Schmid J. "Integrated patient simulation training for the astronaut Crew Medical Officers (CMO) and flight surgeons of STS-118 conducted by the Medical Operation Support Team (MOST)--a preliminary exercise." 79th Annual Scientific Meeting of the Aerospace Medical Association, Boston, MA, May 11-15, 2008. Aviat Space Environ Med. 2008 Mar;79(3):314. , Mar-2008
Abstracts for Journals and Proceedings	Hurst V, Doerr HK, Smart K, Polk J, Schmid J, Baumann D, Johnson-Throop KA. "Development of an 'Emergency Medical Drills' excercise for the NASA Extreme Environment Mission Operation (NEEMO)--a collaboration between the Medical Simulation Laboratory Working Group (MSLWG) and the Medical Operation Support Team (MOST)." 79th Annual Scientific Meeting of the Aerospace Medical Association, Boston, MA, May 11-15, 2008. Aviat Space Environ Med. 2008 Mar;79(3):314. , Mar-2008
Books/Book Chapters	Doerr HD, Murray WB. "How to build a successful simulation strategy: the simulation learning pyramid." in "Clinical Simulation : operations, engineering, and management." Ed. R.R. Kyle, W.B. Murray. Amsterdam ; Boston : Academic Press, 2008., Jun-2008
Significant Media Coverage	Doerr HD. "The five part series 'Mars Rising' covered many aspects of the trip to Mars." Discovery Channel, April 2008., Apr-2008